

Nithin Bobbiligama

DATA ENGINEER

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PROFESSIONAL SUMMARY

- Cloud-Native Data Engineer (Azure & AWS) with 6+ years of experience designing scalable ETL/ELT pipelines, real-time streaming architectures, and enterprise data Lakehouse solutions. Expertise in Azure Data Factory, Databricks, Synapse, Snowflake, and AWS Glue to deliver secure, high-performance analytics at scale.
- Proficient in Python, SQL, and PySpark, with expertise in Apache Spark, Kafka, Flink, Kinesis, Hive, and Sqoop to design reusable ingestion frameworks and real-time streaming pipelines for high-volume data processing.
- Skillful knowledge on modern data platforms such as Snowflake, Synapse, Redshift, Cosmos DB, MongoDB and MySQL, managing both structured and semi-structured data at scale.
- Delivered actionable insights through dashboards built with Power BI, Tableau and Quick Sight, collaborated within Agile/Scrum/Kanban environments to ensure iterative and high-quality delivery.

TECHNICAL SKILLS:

- Programming Languages & Big Data:** Python, Scala, Java, SQL, Apache Spark, PySpark, Hadoop, Hive, Pig, HBase, Sqoop, IBM DataStage, Spark SQL, MapReduce, Apache (Kafka, Flink), Airflow, Oozie.
- Databases & Warehousing:** Azure Synapse, Snowflake, Redshift, MySQL, PostgreSQL, SQL Server, Azure SQL DB, DynamoDB, MongoDB, HBase, Teradata.
- Cloud Platforms & Services:** Azure (ADF, Databricks, Function, Logic Apps), AWS (EMR, Glue, Lambda, EC2, S3, RDS, Redshift, Kinesis, IAM, Elastic search, CloudWatch).
- CI/CD & Version Control:** Azure DevOps, Jenkins, GitHub Actions, Git, GitHub, GitLab, Bitbucket.
- Containerization & Orchestration:** Docker, Kubernetes, AWS EKS.
- Core Competencies & Methodologies:** Custom Data Quality Frameworks, RCA, Governance Policies, Agile/Scrum, Kanban, SDLC.

WORK EXPERIENCE:

Nationwide, Columbus, Ohio, USA.

Aug 2023 – Present

Azure Data Engineer

- Designed and optimized end to end ETL pipelines using Azure Data Factory, Synapse and ODS to accelerate data delivery and support 15+ real-time reporting use cases.
- Built event-driven architectures with Azure Functions, Logic Apps and Kafka, automating data integration workflow and reducing manual efforts and enabling near real-time integrations.
- Engineered high-volume data workflows in Databricks and Spark (PySpark), reducing batch processing time and boosting analytics performance.
- Automated ingestion from diverse sources (HDFS, SQL, flat files) using IBM DataStage and Python frameworks, reducing data latency and improving onboarding.
- Delivered 15+ executive-level dashboards using Power BI, integrated CI/CD with Azure DevOps and Git, and reduced release cycles by 35%.
- Integrated real-time monitoring and alerting using Azure Monitor, Log Analytics and Application Insights, improving system observability and reducing issue resolution time

Humana, Dublin, Ohio, USA

Jul 2022 – May 2023

Data Engineer

- Designed and optimized real-time data pipelines using Apache Spark, PySpark and AWS EMR to analyze provider performance, enabling faster decision-making across healthcare operations.
- Developed and maintained data ingestion workflows integrating RESTful APIs using Python, Capturing real-time data provider and claims data for analytical dashboards.
- Reduced processing time and infrastructure costs by tuning Redshift, implementing auto-scaling on EMR and designing scalable storage with MySQL and DynamoDB.
- Streamlined ingestion and live analytics by integrating AWS Kinesis with RESTful APIs, supporting low-latency dashboards and improving provider insight visibility.
- Delivered ETL pipelines using IBM DataStage, Snowflake, and Spark SQL, while implementing CI/CD automation with Jenkins and GitHub, cutting deployment time by 30%.
- Improved data quality and validation by implementing robust error handling, schema checks and audit frameworks, ensuring reliability across multiple healthcare data sources.

Experian, Hyderabad, India

Oct 2019 – Jun 2021

Data Engineer

- Developed scalable ETL pipelines using Sqoop, Python and Shell scripts to ingest data from SQL Server, MySQL and flat files into HDFS, supporting batch data workflows.
- Reduced ETL effort by 40% through automation of multi-source integration with IBM DataStage and AWS Glue and optimized data transformation using Apache Spark and PySpark.
- Built and deployed real-time streaming pipelines with AWS Kinesis, Kafka and Redshift, improving SLA adherence and delivering near-instant insights for business operations.
- Managed diverse data models using MySQL, MongoDB and HBase and enabled real-time analytics via Tableau, Snowflake and S3 dashboards.
- Implemented and Orchestrated jobs with Apache Airflow & Oozie, ensuring scalable, fault-tolerant pipelines with CI/CD automation.

Novartis, Hyderabad, India

Mar 2018 - Sep 2019

Program Analyst

- Automated data extraction and transformation using Python, improving data accuracy and reducing manual intervention across MySQL and MongoDB data sources.
- Built scalable ETL pipelines with Sqoop, Hive, Pig and Oozie on Hadoop and recommended Apache Airflow for modernize workflow orchestration.
- Developed a custom Python GUI tool for hardware simulation, enabling early QA testing without physical devices and accelerating test cycles.
- Optimized Spark SQL performance and supported interactive data visualizations using JavaScript, AJAX and JSON for business insights.
- Designed and scheduled reporting pipelines, integrating structured and semi-structured data and ensuring consistent delivery through automated workflows.

EDUCATION:

- University of Dayton
Master of Science in Computer Science.

CERTIFICATIONS:

- Azure Associate Data Engineer.
- Power BI Data Analyst Associate.